```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder,StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error
```

```
df =sns.load_dataset('tips')
print(df.head())
```

```
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4
```

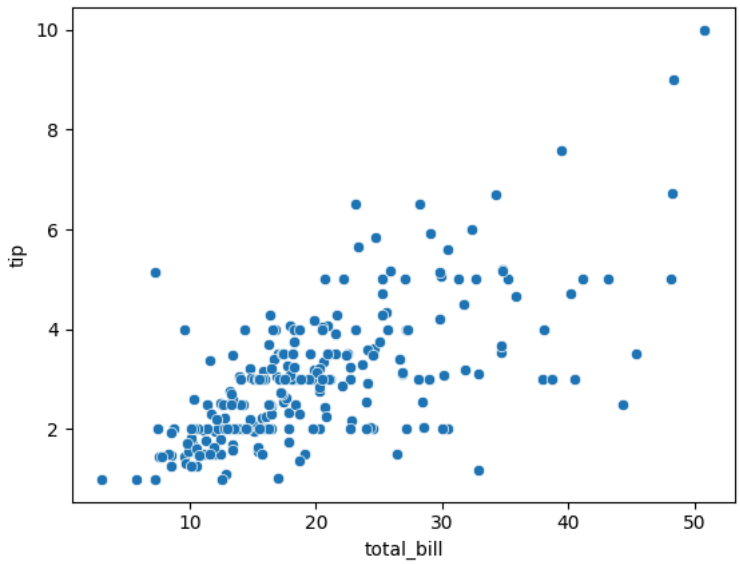
```
x=df.drop('tip',axis=1)
y=df['tip']
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   total_bill  244 non-null    float64
 1   tip         244 non-null    float64
 2   sex         244 non-null    category
 3   smoker      244 non-null    category
 4   day         244 non-null    category
 5   time        244 non-null    category
 6   size        244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

	0
total_bill	0
tip	0
sex	0
smoker	0
day	0
time	0
size	0

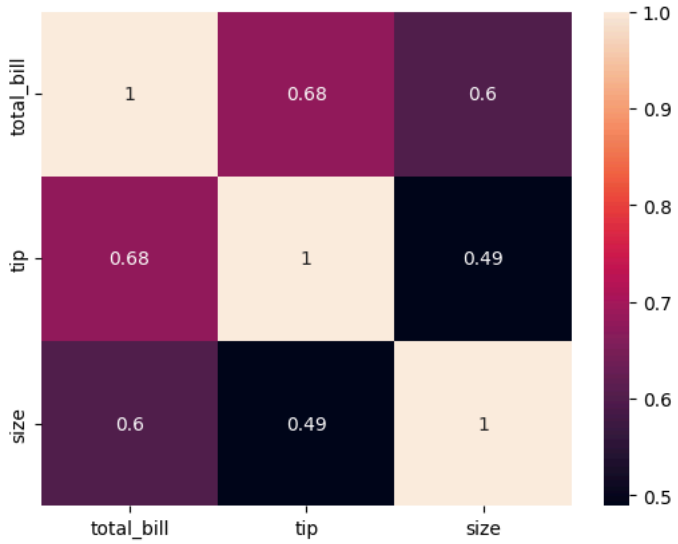
dtype: int64

```
sns.scatterplot(x='total_bill', y='tip', data=df)
plt.show()
```




```
sns.heatmap(df.corr(numeric_only=True), annot=True)
```

```
<Axes: >
```




```
cat_cols = ['sex', 'smoker', 'day', 'time']
num_cols = ['total_bill', 'size']

features = ['total_bill', 'size', 'sex', 'smoker', 'day', 'time']
X = df[features]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

preprocessor = ColumnTransformer(
    [('cat', OneHotEncoder(drop='first'), cat_cols),
     ('num', StandardScaler(), num_cols)]
)

pipeline = Pipeline([
    ('preprocessor', preprocessor),
    ('model', LinearRegression())
])
```

```
pipeline.fit(X_train, y_train)
```

```
Pipeline
preprocessor: ColumnTransformer
cat
OneHotEncoder
num
StandardScaler
LinearRegression
```

```
y_pred = pipeline.predict(X_test)
print("R2 Score:", r2_score(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
```

```
R2 Score: 0.43730181943482505
RMSE: 0.838663580790063
```

```
import pickle
```